Kinematics and Dynamics of a Vehicle Model

Vehicle Dynamics and Control for Mobility
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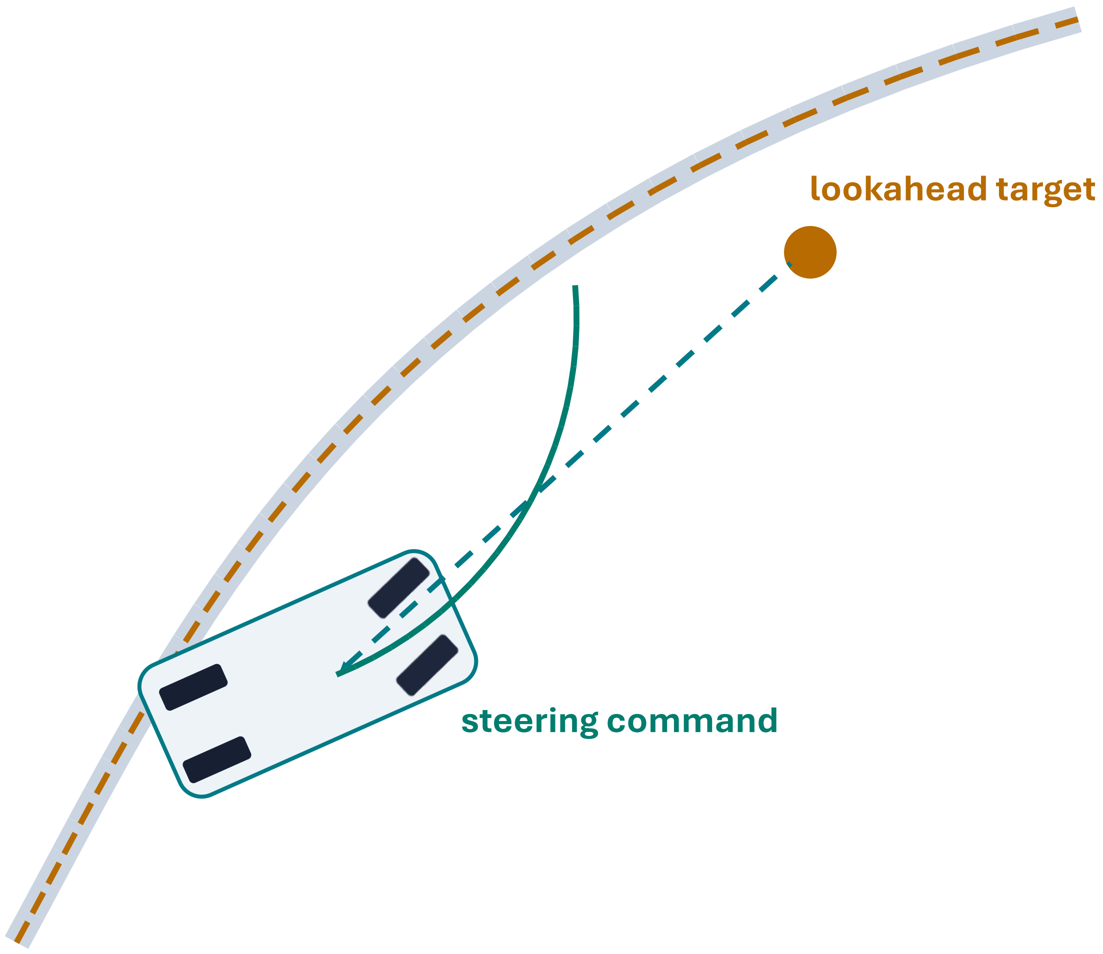
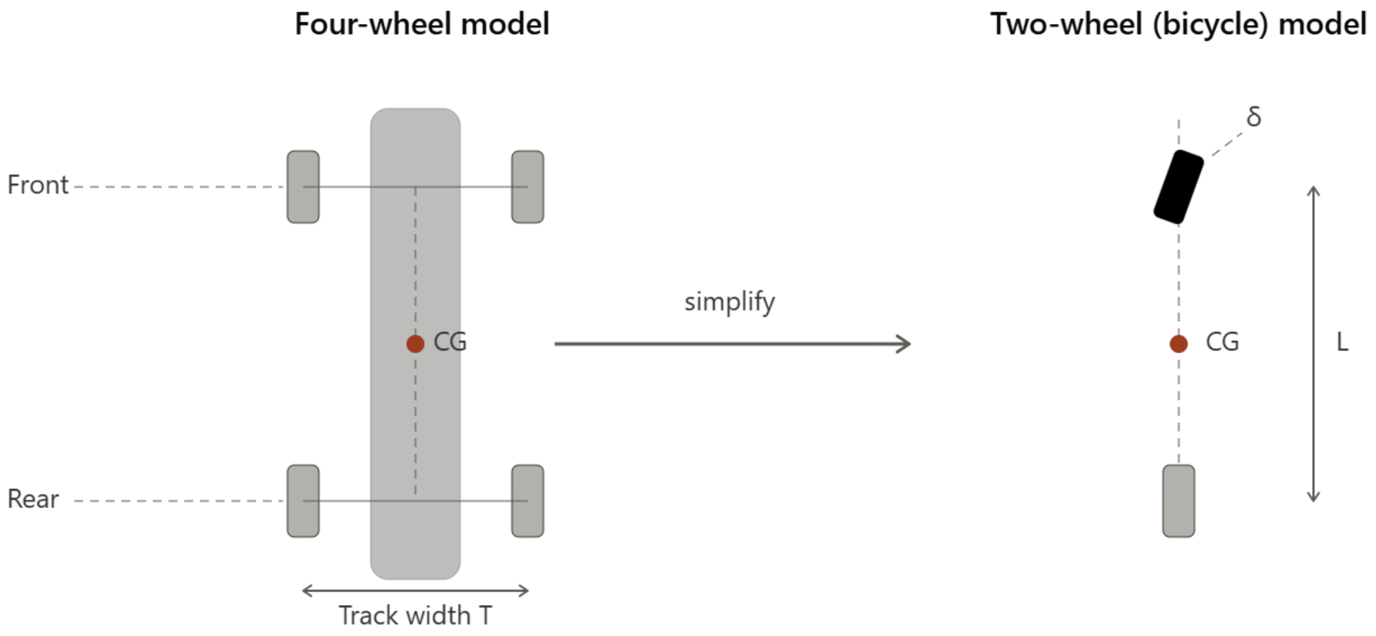
From geometry to steering to control (path tracking)

The bicycle model and the pure pursuit and model predictive control algorithms



A Simplified Vehicle Model

- Bicycle Model
  - Consolidate four wheels into two wheels “like” a bicycle
  - Neglect vehicle pitching and rolling motions
  - Consider only motions in the plane; lateral vehicle motion



A Simplified Vehicle Model

- Review key kinematics concepts
- Develop basic kinematic relations
- Tire force model
- Governing equations of motion for control

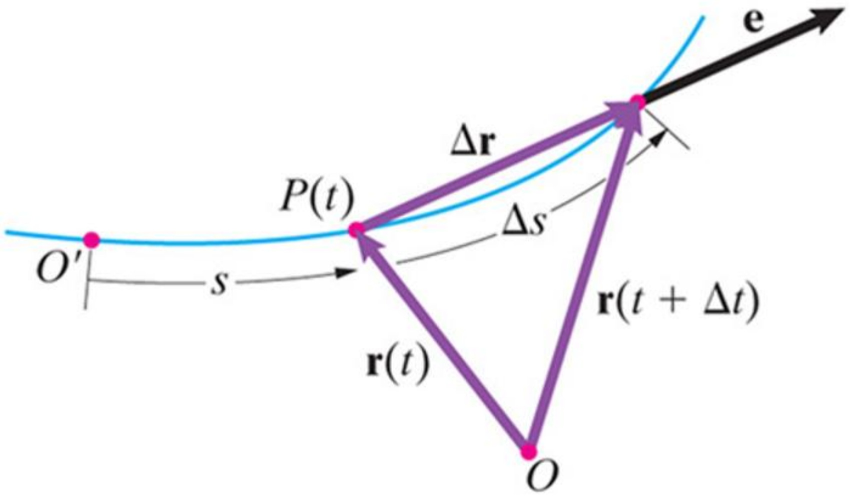
An Important Kinematic Concept

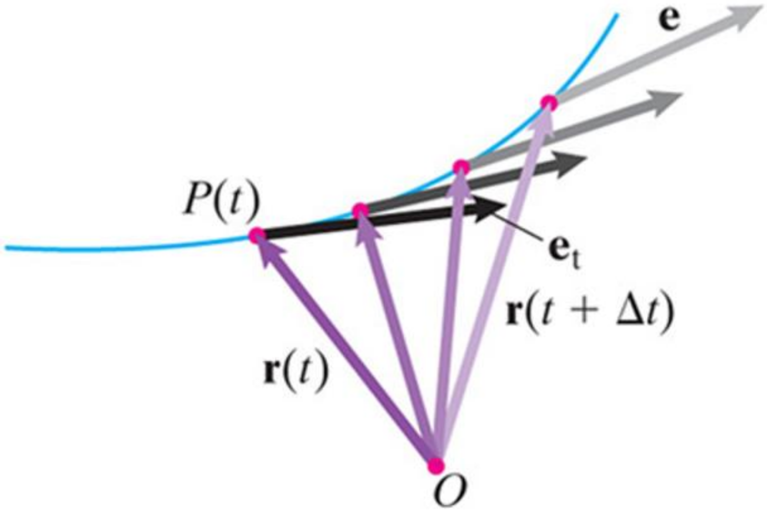
Definition of the velocity vector of a point:

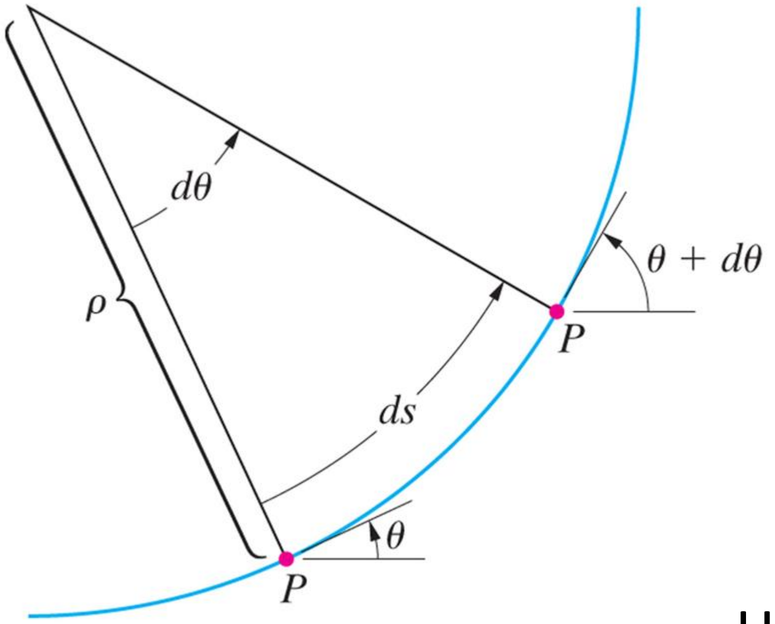
$$\mathbf{v} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t} \quad \left(= \frac{D\mathbf{r}}{Dt} \right)$$

$$= \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}\, \mathbf{e}_t = v\, \mathbf{e}_t = \dot{s}\, \mathbf{e}_t \quad (\mathbf{e}_t \text{ is the unit tangential vector})$$

Velocity vector of a point is tangential to its path/trajectory







Alternate expression for the velocity vector in terms of the path distance parameter s:

$$\boxed{\mathbf{v} = v\,\mathbf{e}_t = \dot{s}\,\mathbf{e}_t = \rho\frac{d\theta}{dt}\,\mathbf{e}_t}$$

Here, ρ is the <u>instantaneous</u> radius of curvature

For planar motions: $\rho = \dfrac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}$

The above formula can be derived from geometry.

Time derivative of velocity vector with respect to an inertial reference frame $\frac{D}{Dt}$:

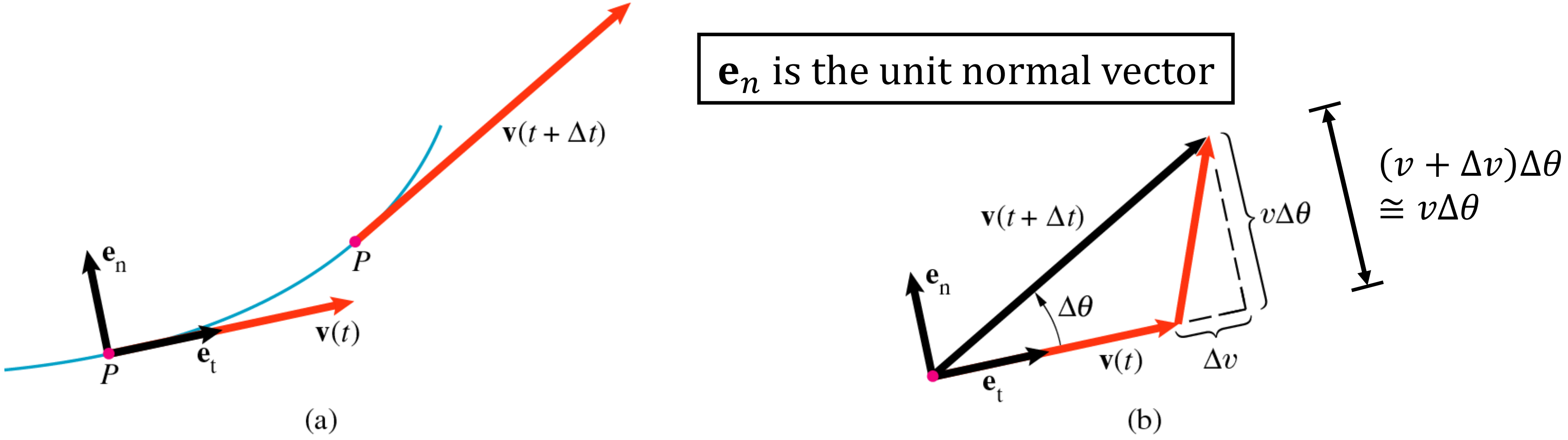
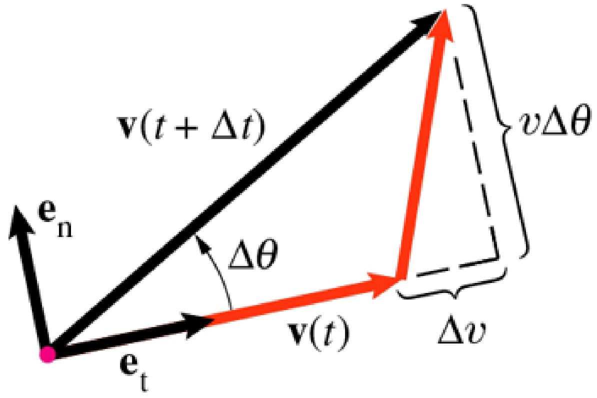


Figure 13.27

(a) Velocity of P at t and at $t + \Delta t$.
(b) The tangential and normal components of the change in the velocity.

$$\boxed{\mathbf{a} = \frac{D\mathbf{v}}{Dt} = \frac{D(v\,\mathbf{e}_t)}{Dt} = \frac{dv}{dt}\mathbf{e}_t + v\frac{D\mathbf{e}_t}{Dt} = \frac{dv}{dt}\mathbf{e}_t + v\frac{d\theta}{dt}\mathbf{e}_n\,, \qquad \text{where } \mathbf{e}_n = \frac{D\mathbf{e}_t}{Dt}\Big/\left|\frac{D\mathbf{e}_t}{Dt}\right|}$$

$$\boxed{\mathbf{a} = \frac{D\mathbf{v}}{Dt} = \dot{v}\,\mathbf{e}_t + v\frac{d\theta}{dt}\mathbf{e}_n = \dot{v}\,\mathbf{e}_t + \frac{v^2}{\rho}\,\mathbf{e}_n}$$



$$a_t = \dot{v}$$

$$a_n = \frac{v^2}{\rho}$$

tangential acceleration due to change in speed

Normal acceleration due to change in direction of velocity

This formula for the acceleration vector is also valid for 3D motions

Example: Circular motions

$$\rho = R, v = \dot{s} = \frac{d}{dt}(R\theta) = R\omega \;\Rightarrow\; \mathbf{v} = R\omega\,\mathbf{e}_\mathrm{t}$$

Velocity vector is tangential to the circular path.

$$a_t = \dot{v} = R\frac{d\omega}{dt} = R\alpha, \alpha = \text{angular acceleration,}$$

$$a_n = \frac{v^2}{\rho} = R\omega^2$$

$$\mathbf{a} = R\alpha\,\mathbf{e}_\mathrm{t} + R\omega^2\,\mathbf{e}_\mathrm{n}$$

Using the dot product technique, we can obtain the relations between the coordinate systems:

$$\mathbf{e}_\mathrm{t} = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$$

$$\mathbf{e}_\mathrm{n} = -\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}$$

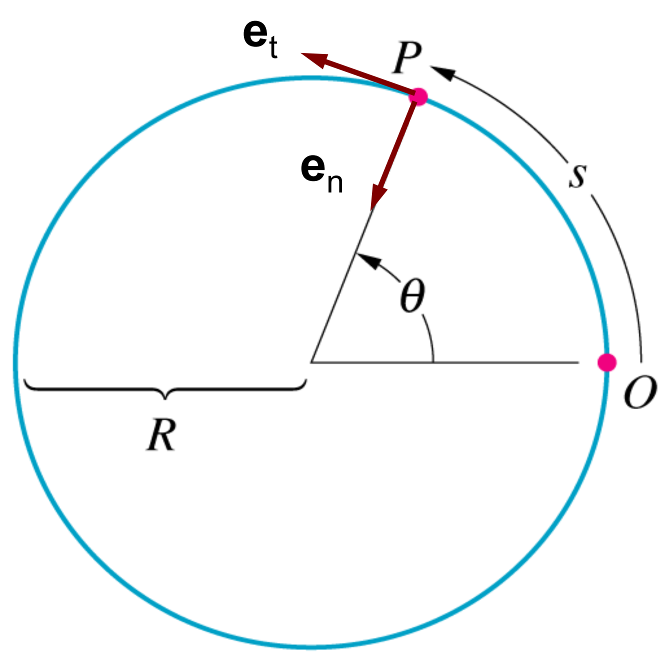


Figure 13.32

A point moving in a circular path.

Kinematic Model of Lateral Motion

- O is the instantaneous center of rotation (ICR)
- Assume all points on the "bicycle" are rotating about O
- C is the center of gravity
- δ_f is steering angle of the front tire
- δ_r is steering angle of the rear tire; $\delta_r = 0$ for front wheel drive
- β is the side-slip angle
- ψ is the yaw angle with respect to X-Y inertial (fixed) coordinate system

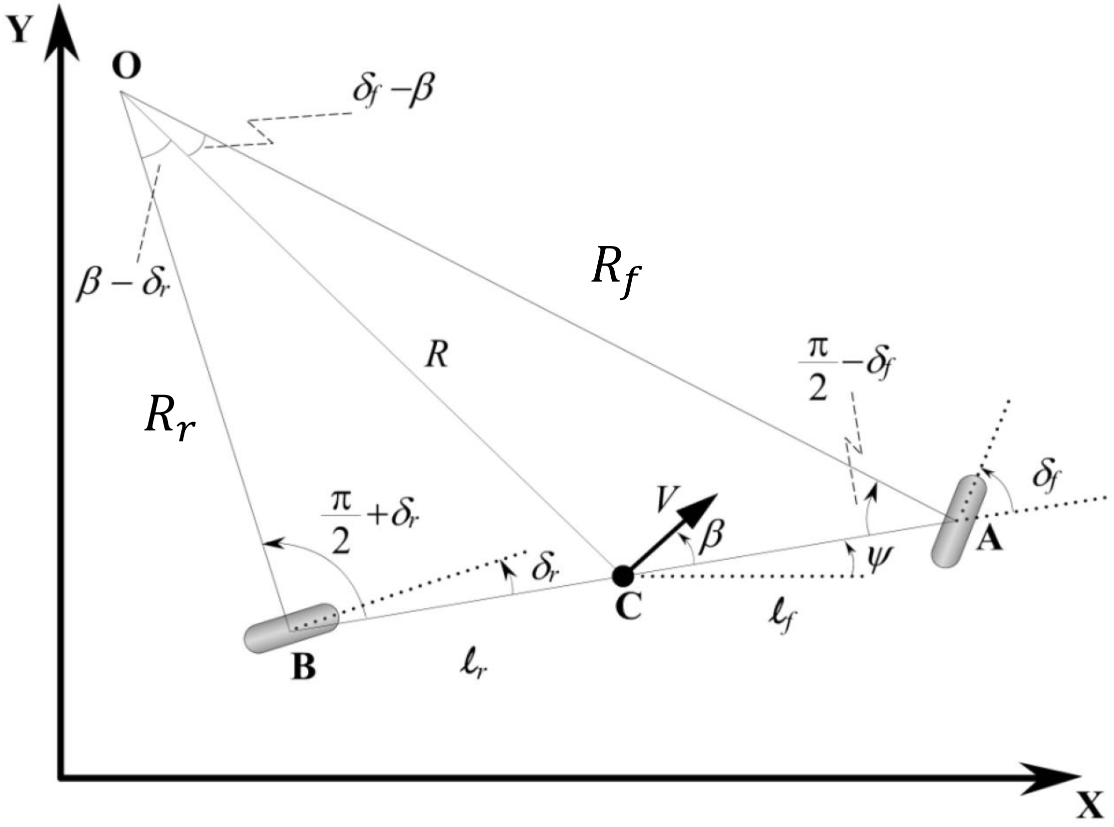


Figure 2-3. Kinematics of lateral vehicle motion

Kinematic Model of Lateral Motion

- In this model, assume zero slip angle at the front and rear wheels, i.e., the velocity vectors are in the direction of the steering.
- To simplify analysis, assume $\delta_r = 0$.

Assume V is constant, thus

$$\dot{X} = V\cos(\psi + \beta)$$

$$\dot{Y} = V\sin(\psi + \beta)$$

Also, define angular velocity of the body

$$\dot{\psi} = \omega = \frac{V}{R}$$

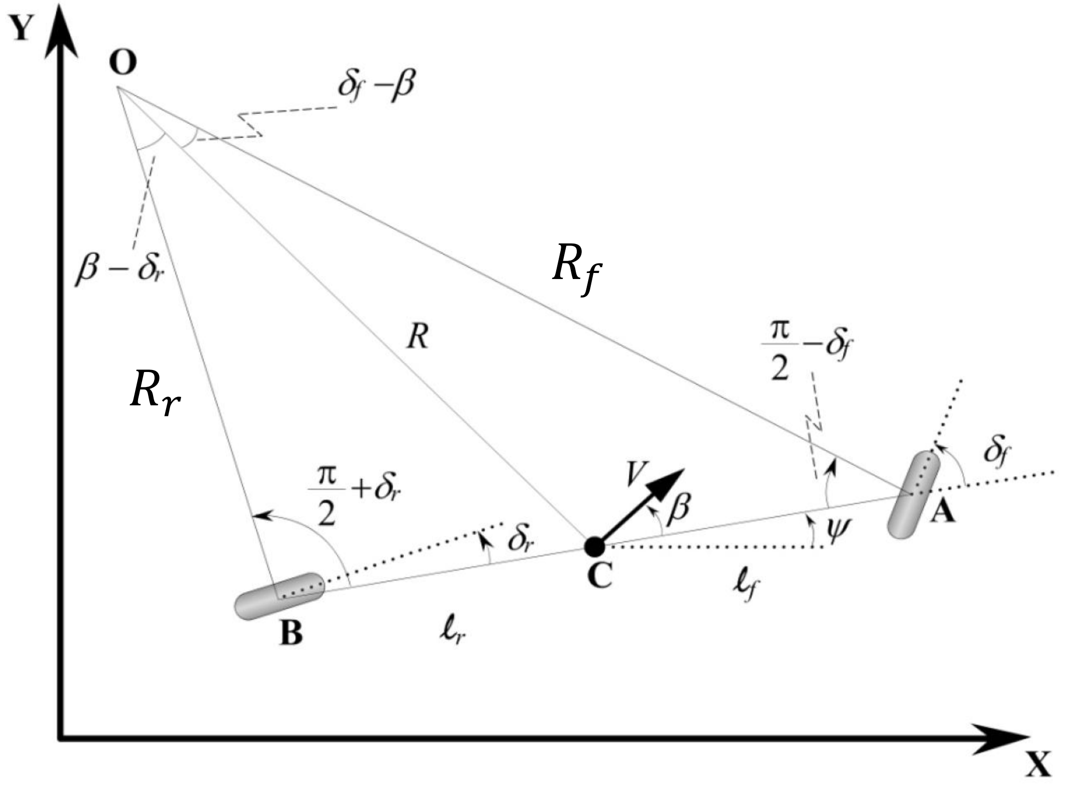


Figure 2-3. Kinematics of lateral vehicle motion

Kinematic Model of Lateral Motion

From geometry, assume $\delta_r = 0$, we have

$$\tan\delta_f = \frac{\ell_r + \ell_f}{R_r}$$

We also have

$$\tan\beta = \frac{\ell_r}{R_r} = \frac{\ell_r}{\ell_r + \ell_f}\tan\delta_f$$

Thus, we have found a relation between β and the input steering angle δ_f. From geometry, we have

$$\cos\beta = \frac{R_r}{R} \Rightarrow \frac{1}{R} = \frac{1}{\ell_r + \ell_f}\tan\delta_f\cos\beta$$

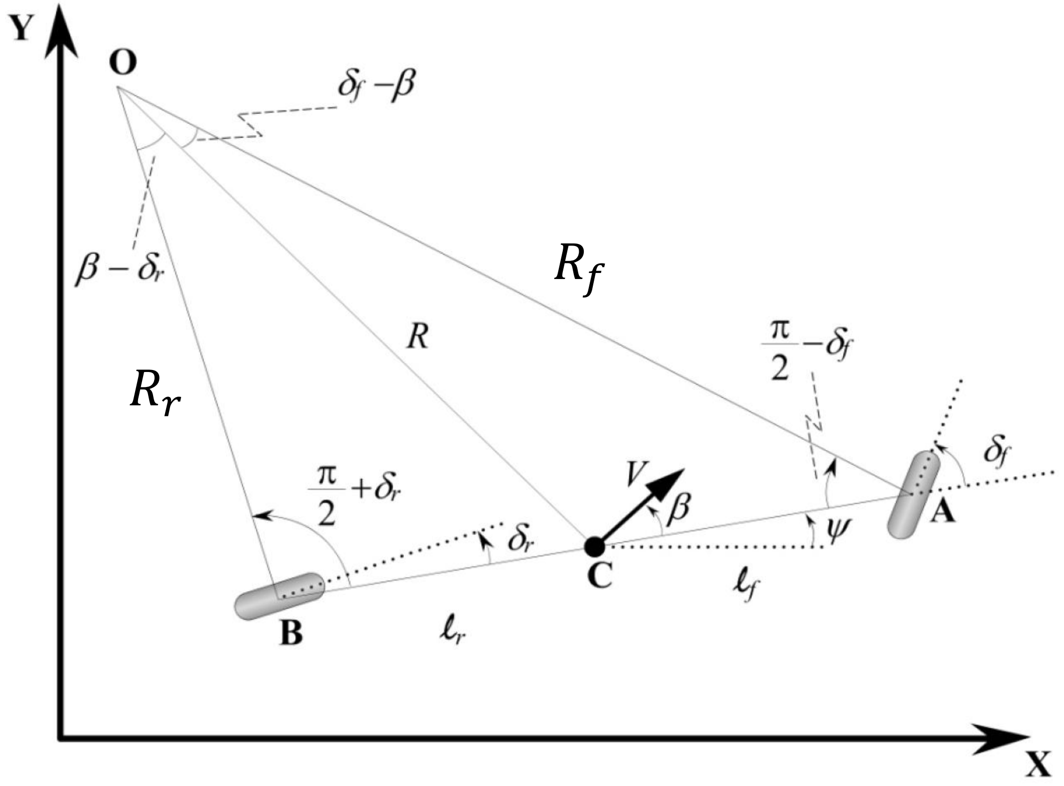


Figure 2-3. Kinematics of lateral vehicle motion

Kinematic Model of Lateral Motion

In summary, we have obtained three coupled nonlinear ordinary differential equations to describe the time evolution of the three kinematic variables (X, Y, ψ) or state variables as a function of the input variable δ_f:

$$\dot{X} = V\cos(\psi + \beta)$$

$$\dot{Y} = V\sin(\psi + \beta)$$

$$\dot{\psi} = \omega = \frac{V}{\ell_r + \ell_f}\tan\delta_f \cos\beta$$

where, $\beta = \tan^{-1}\left(\frac{\ell_r}{\ell_r+\ell_f}\tan\delta_f\right)$.

Kinematic Model of Lateral Motion

If the rear wheel steering angle is non-zero, $\delta_r \neq 0$, it can be shown the differential equations can be expressed in terms of the two input variables of steering angles δ_f and δ_r:

$$\dot{X} = V\cos(\psi + \beta)$$

$$\dot{Y} = V\sin(\psi + \beta)$$

$$\dot{\psi} = \omega = \frac{V\cos\beta}{\ell_r + \ell_f}(\tan\delta_f - \tan\delta_r)$$

where, $\beta = \tan^{-1}\left(\frac{\ell_r \tan\delta_f + \ell_f \tan\delta_r}{\ell_r + \ell_f}\right)$. The kinematic equations can be further extended to include the case of non-constant V.